



Article

'George's Gymnastics Program', a Community Gymnastic Program for Children with a Disability: Perceptions of Participant Stakeholders

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Abstract: *Background:* The aim of this study is to understand what children with a disability like and dislike about participating in a gymnastics program, and to understand the views of the adults supporting the program. *Methods:* Children with a mild intellectual disability (aged 5–12) from a primary school providing special education participated in one of four gymnastics programs (1 h/week for 5 weeks) at a gymnastics centre. The program included ~25–50 children per group, with ~20 adults providing support, including gymnastics coaches, teachers, and adult students on placement. To evaluate the program, children (11–12-year-olds; $n = 51$) were invited to complete a written two-page reflection, and the adults were invited to participate in one of three focus groups. *Results:* 150 children attended one of four programs. Twenty-six 11–12-year-olds participated in the evaluation ($n = 26/51$, 51%), and, from the written responses, the themes of 'individual preference' and 'resilience' emerged. It was found that 96% ($n = 25/26$) reported they would like to do more gymnastics, 88% ($n = 23/26$) reported positive feelings during gymnastics, and 65% ($n = 17/26$) expressed concerns such as feeling tired or hot. Twenty-three adults participated in a focus group. Key themes included the positive impact the program delivery had on the children's participation; and the positive impact on coaches, students, and teacher's disability attitudes and knowledge. *Conclusions:* This gymnastics program was well-liked by children with a disability. To optimise participation, promoting individual choice and control may be of benefit.

Keywords: disability; physical activity; children; school; education



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1. Introduction

People of all ages living with a disability are less likely to meet physical activity guidelines, compared to people living without a disability, and this contributes to serious inactivity-related health problems [1]. For children and adolescents with a disability, less than half meet physical activity guidelines, and, while inclusion in physical activity programs optimises physical function and improves well-being, attendance is limited and

this contributes to lower fitness levels and higher obesity levels when compared to children without a disability [2–4].

The World Health Organization states that environmental factors ‘make up the physical, social and attitudinal environment in which people live and conduct their lives’, and these environmental factors can present as a barriers or a facilitator to a person’s function [5]. Barriers to physical activity participation for children with a disability include practical limitations such as program and staff capacity, time constraints, the child’s preferences, parental behaviour, an unwillingness to be inclusive, negative attitudes towards disability, and a lack of skilled instructors and opportunities [6–8]. Modifiable barriers to physical activity, such as the physical environment, can often be altered to positively impact physical activity participation, whereas non-modifiable factors associated with low physical activity, such as intellectual ability and parents’ ethnicity, cannot be altered [9]. This is in contrast to the facilitators to physical activity participation for children with a disability which include the child’s desire to be active, the involvement of peers, family support, programs that promote success and inclusion through fun and inclusive activities, programs that have skilled staff and a pathway as children grow and develop their skills, and programs with strong partnerships across the disability, sport, education, and government sectors [6–8]. In addition, while some research examining inclusive physical activity programs focus on the voice of the participating child with a disability, such as that led by Shields and colleagues [7], most commonly, the literature will focus on the barriers, facilitators, and views reported by the adults who support such programs [10].

The sport of gymnastics, that includes floor, beam, bars, vault, and trampoline, is an example of a community sports program that focuses on physical activity, and it is reported that an inclusive gymnastics program can directly benefit children with a disability [11], for example, physical benefits, as well as benefits to confidence, self-esteem, and social development [12]. While inclusivity for children with a disability can vary from sport to sport, for gymnastics, inclusivity requires an accessible environment, an inclusive club culture, recognition and enjoyment, and knowledgeable coaches to facilitate the children’s participation in gymnastics [13]. Access to gymnastics is limited for children with a disability, and this has been exacerbated by coaching shortages in the post-COVID-19 era, where youth sporting coaches have found it difficult to re-engage in their role as coach [14].

The current study focuses on a community gymnastics program for children with a disability. For this evaluation of attitudes, the primary aim was to understand what children with a disability (aged 11–12 years) like and dislike about participating in an inclusive community gymnastics program; and the secondary aim was to understand the impact, barriers, and facilitators of an inclusive community gymnastics program from the perspective of participating gymnastics coaches, adult students on placement supporting the program, and teachers.

2. Materials and Methods

Methodologically, this research was grounded in the principles of inclusive disability research [15]. That is, the voices of all stakeholders were valued, methods which elicit diverse voices are sought and used [16], and, where feasible, stakeholders are invited to be included in the machinery of research production including data analysis, writing, authoring, and presenting [17].

This observational study was reported in accordance with the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) [18] guidelines and was approved by the Monash University Human Research Ethics Committee (Project ID #38902). This study was conducted in the state of Victoria, Australia and comprised a partnership across six sectors: disability (primary-school-aged children with a disability), sport (Waverley Gymnastics Centre), education (Vermont South Special School), government (Victorian Government funding), the tertiary education sector (Monash University), and the ‘technical and further education’ or TAFE sector (Holmesglen Institute).

2.1. Setting

In 2022, Waverley Gymnastics Centre, in partnership with Vermont South Special School and Monash University, was awarded AUD \$50,000 via a Victorian Government JumpStart! grant with the intent of providing a time-limited inclusive gymnastics program for children with a disability. The gymnastics program was delivered between February and November 2023, at the Waverley Gymnastics Centre in Victoria (Australia). Waverley Gymnastics Centre is a world-class, not-for-profit community-based gymnastics club that provides multiple gymnastics programs including junior gymnastics, educational gymnastics, and competitive squad gymnastics, and, in 2023, the centre had about 2000 gymnast members.

The gymnastics program participants were children from Vermont South Special School in Victoria (Australia). Vermont South Special School is a public primary school for children with a mild intellectual disability as defined by the child having an intelligence quotient (IQ) between 50 and 70, and this present as delays to cognitive development and delays to social and daily living skills. The children may also have additional types of disability such as autism, low vision, reduced mobility, and behaviours of concern. In 2023, during the study, the school had four sections: Foundation (children aged 5–6, $n = 25$); Junior (children aged 7–8, $n = 35$); Middle (children aged 9–10; $n = 39$); and Senior (children aged 11–12; $n = 51$).

2.2. Intervention

As a part of the grant activity, all four sections of Vermont South Special School participated in a 5-week gymnastics program ('George's Gymnastics Program') within school hours, as a part of the planned usual school activities for 2023. This gymnastic program was a new program and it was specifically designed for children with disabilities. The design process for the program was iterative and included all authors co-designing some or all of the program. The year prior the program was piloted on a single occasion with 40 children with a disability aged 8–9; however, the program itself had not been validated.

One section of the school completing the program per school term (noting that, from January to December, there are four 10-week terms per year in Victorian schools). Participation in the program came at no cost to the child or the school. Providing support and direction for the children during each session was a minimum of 22 adults, including four gymnastics coaches employed by Waverley Gymnastics Centre, eight teaching staff employed by Vermont South Special School, eight adult students on placement from Monash University (physiotherapy, occupational therapy, and teaching) and/or Holmesglen TAFE (community service and disability support work), as well as a program manager employed by Waverley Gymnastics Centre (qualified school teacher) and program mentor employed by Monash University (qualified physiotherapist).

2.3. Participants

There were two groups of participants for this study: (i) children who were senior students (aged 11–12 years) from Vermont South Special School participating in the gymnastic program, and (ii) adults supporting the gymnastic program, which included gymnastic coaches, students from university or TAFE on placement, and teachers.

During each session of the inclusive community gymnastics program, between 25 (Foundation section) and 51 children (Senior section) attended, with a total of 150 children attending the 5-week program, across all four sections of the school. While the gymnastics program was provided to all children from Vermont South Special School, the evaluation from the children's perspective only included the children (aged 11–12 years) enrolled in the senior section of the Vermont South Special School in 2023 (group i). These children were selected as informants as they were more likely to be able to provide feedback on the programs compared to the younger children. There were no exclusion criteria for this group. Within the week prior to the senior students completing the data collection tool (a 2-page post-program worksheet), a hard copy of the study participant information and consent form was placed in each Senior child's school diary, that was then placed in the child's

school bag, and delivered to the parent/guardian. While all Senior children completed the worksheet, only children whose parent/guardian had provided written informed consent had their worksheet provided to the research team and were included in the evaluation. Regarding the sample size, the sample was one of convenience, and there were no *a priori* sample size calculations.

The second group of participants involved adults supporting the gymnastic program (i.e., gymnastic coaches, university or TAFE students on placement, and teachers) and each of the groups were evaluated separately. For the focus groups (group ii), seven coaches, eight university or TAFE students on placement, and eight teachers participated. Inclusion and exclusion criteria were consistent across the three sub-groups of adults. For inclusion, they needed to have worked in the inclusive community gymnastics program on one or more occasions during 2023 and were aged 18+ years. Exclusion criteria included being under 18 years and not having exposure to George's Gymnastics Program in 2023.

Coaches (aged 18+ years) were invited to participate in the coaches' focus group. To participate in the focus group, the gymnastics program mentor emailed the coaches, where the voluntary nature of participation was highlighted. Prior to participation, 'coaches' provided written informed consent. Regarding the sample size, the sample was one of convenience, and, *a priori*, the focus group aimed to recruit >6 participants for the 'coaches' focus group.

Adult students (aged 18+) on placement from Monash University (physiotherapy, occupational therapy, and teaching) and Holmesglen TAFE (Technical and Future Education) (community service and disability support work) were invited to participate in the 'students on placement' focus group. For context, TAFE and universities are both educational institutions, with TAFE often focusing on practical vocational courses, while universities offer more traditionally academic courses. To participate in the focus group, the gymnastics program mentor emailed the students, where the voluntary nature of participation was highlighted. Prior to participation, students provided written informed consent. Regarding the sample size, the sample was one of convenience, and, *a priori*, the focus group aimed to recruit >6 participants for the 'students' focus groups.

Teachers (aged 18+) from Vermont South Special School were invited to participate in the coaches' focus group. To participate in the focus group, the school Principal emailed the teachers, where the voluntary nature of participation was highlighted. Prior to participation, teachers provided written informed consent. Regarding the sample size, the sample was one of convenience, and, *a priori*, the focus group aimed to recruit >6 participants.

For both the children and adult participant groups, research participants could withdraw from the research project up until the point of data de-identification. There was no research participant follow-up.

2.4. Outcome Measures—Children—Group i

Procedure: To understand what children with a disability like and dislike about participating in an inclusive community gymnastics program, participants completed a two-page post-program worksheet. Pre-testing of the two-page worksheet was completed by one student in the Junior section at Vermont South Special School (10 years of age), under the guidance of the first author (NB). As the Junior section student was able to complete the questions and drawing with minimal prompts, no modifications were made. Participants completed a 2-page post-program worksheet during class time, under the supervision of the English Specialist teacher who had not been involved in the gymnastics program, to reduce bias and ensure consistency of instructions across the four senior classes participating in the program.

Tool: The worksheet asked four questions (i) what did you like about gymnastics? (ii) what didn't you like about gymnastics? (iii) how did you feel during gymnastics? and (iv) would you like to do more gymnastics? The worksheet also asked the children to draw themselves at gymnastics. The worksheet was presented on an A3 page, divided in half, to

make a 2-page worksheet. The evaluation worksheet was completed in August 2023, and this was completed the week after the 5-week program.

2.5. Outcome Measures—Adults—Group ii

Three focus groups were conducted with adult participants supporting the gymnastic program (i.e., gymnastic coaches, students on placement, and teachers). Focus groups for the coaches and students on placement were conducted in August 2023 and the focus group including teachers was conducted in September 2023. Focus groups were moderated by author NB (female, physiotherapist and health economist, and experienced researcher), and took between 45 and 60 min to complete. For each of the focus groups (coach, student, and teacher), the topic guide was grouped into ‘impact’ topics and ‘local environment model’ [19] topics, with details presented in Appendix A. The impact topics were based on perceived self-impact personally or professionally. Selected as a model which incorporates components of the World Health Organization’s International Classification of Functioning, Disability, and Health (ICF) framework [20], the local environment model [19] focuses upon the mechanisms through which context influences participation as an outcome. For example, the local environment model [19] topics of body function and learning aligned to body functions and structures of the ICF [20]. Considering activities and participation of the ICF, aligned topics in the local environment model [19] included enjoyment in physical activity, independence and choice. Aligned with the natural environment and human-made changes to environment within the ICF [20] were local environment model [19] topics of time use and safety. For support and relationships in the ICF [20], topics of social connections, family-centredness, friends, and specialised health professionals were aligned from the local environment model [19]. The ICF acknowledges personal factors but does not seek to categorise or define them [20]. Personal factors are however included in the local environment model [19] and include fun, achievement, and aspiration.

2.6. Data Analysis

The written participant post-program worksheet data were firstly described for each question. Inductive content analysis was then completed by one author (NB) for all questions [21]. Inductive content analysis was chosen because it is well-suited to small-scale health-based research, there were no previous studies dealing with this exact topic, and the research team did not have a list of pre-determined content items [21]. To understand the views, barriers, and facilitators, and impact of an inclusive community gymnastics programs from the perspective of gymnastics coaches, university/TAFE students on placement, and teachers, the focus group data also underwent an inductive content analysis [21]. This involved all authors reviewing of the transcripts, engaging in subsequent discussion, and supporting the inductive content analysis [21]. For each of the focus groups, five to eight members of the authorship team reviewed the transcript and developed individual themes. These individual themes were then brought together through group discussion to reach a consensus.

2.7. Data Integrity

Regarding integrity and trustworthiness, the first author (NB), who facilitated the focus groups, was able to enhance the integrity and trustworthiness of the information provided by building early rapport with the focus group participants (i.e., coaches, the students, and the teachers from the special school). Rapport building for coaches was supported by their history of coaching gymnastics for children with and without a disability over a 5-year period (1995–2000), rapport building for students was supported by their experience as a placement clinical educator across the university and TAFE settings (2000–2017), and rapport building for teachers was supported by their membership on the special school council (2021–2024) [22]. Consistent with inclusive research principles, the involvement of stakeholders who wished to do so in data analysis and manuscript edits, and as co-authors

also provided multiple opportunities to explore and refine the data, building integrity and trustworthiness [23].

3. Results

3.1. Participants

Across the five weeks, the senior student week-to-week attendance of the program was 100%. While all senior students attended the program ($n = 51$), it was the consent rate from the parents/carers that limited the inclusion in the evaluation (51%, $n = 26/51$). Of the 51 Senior children who participated in the program, 26 children ($n = 26/51$, 51%) had a parent/guardian provide written informed consent to participate in the evaluation, and were present at school on the day the worksheet was completed (group i).

For the focus groups (group ii), the coaches were aged between 20 and 50, and were all female, and six were current gymnastics coaches, with one in management (past gymnastics coach). Students were aged between 18 and 40; seven were female, and four were studying community services, three were studying physiotherapy, and one was studying disability support. Teachers were aged between 24 and 55; seven were female, and four were teachers, and four were teacher's aids.

3.2. Children's Experience of the Gymnastic Program (Group i)

Of the children included in the evaluation, 53% ($n = 14/26$) were male, and 96% ($n = 25/26$) reported that they would like to do more gymnastics (Table 1). The themes are summarised in Figure 1 and examples of the children's responses are provided in Figure 2.

Question 1: What did you like about the gymnastics? The equipment and how they felt.

All 26 children ($n = 26/26$, 100%) identified something they liked about the gymnastics program. Children reported they liked the gymnastics equipment (Table 1). Children also reported that they liked the movement associated with using the gymnastics equipment ($n = 12/26$, 46%), for example, "I like jumping", "I like the monkey bar because I could bounce", "I like running fast on the long trampoline", "I liked the trampoline because it made me jump so high", "I did a cartwheel off from the beam", and "I liked the monkey bars because they move". Children also expressed liking the feelings associated with using the gymnastics equipment, for example, "I liked the monkey bars because it was a bit scary", "I liked being upside down", and "I liked gymnastics because my mum came to watch". Finally, one child ($n = 1/26$, 4%) reported they loved multiple parts of the program as it was both "easy and hard".

Table 1. What children liked, disliked, and felt during gymnastics.

Question		Response	Response Rate
1. Liked during gymnastics		Answer provided	$n = 26/26$, 100%
		Equipment	$n = 26/26$, 100%
		Trampoline	$n = 15/26$, 58%
		Bars	$n = 8/26$, 31%
		Rings	$n = 3/26$, 12%
		Moving	$n = 12/26$, 46%
2. Disliked during gymnastics		Answer provided	$n = 20/26$, 76%
		Equipment	$n = 4/26$, 15%
		Moving	$n = 7/26$, 27%
3. Feeling during gymnastics (multiple responses per child)		Answer provided	$n = 23/26$, 88%
		Happy/excited	$n = 23/26$, 88%
		Tired/hot	$n = 6/26$, 23%
		Nervous/bored	$n = 3/26$, 12%

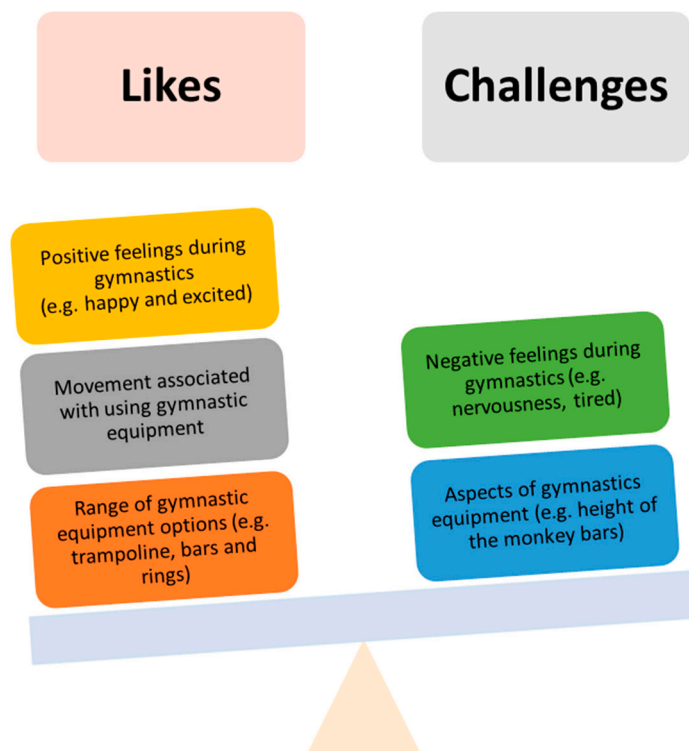


Figure 1. What children liked and found challenging about the gymnastics program.

Question 2: What didn't you like about the gymnastics? The equipment and how they felt.

Twenty children ($n = 20/26$, 76%) identified something they did not like about the gymnastics program. Four children ($n = 4/26$, 15%) noted that they did not like particular aspects of the gymnastics equipment, specifically "I didn't like the monkey bars because it was too high", "I didn't like the bars because the bars was moving", "I didn't like the floor because it was too bouncy", and "I didn't like the beam because it was very high". Seven children ($n = 7/26$, 27%) noted that they did not like how they felt on particular gymnastics equipment, specifically "Backward roll because my tummy was funny", "Beam because my knee was hurting and I was still trying", "Floor because it was not fun", "Bars because it is scary being upside down", "nervous", "I did not like the rainbow [colourful arch to walk across] as I wanted to sleep on it", "It was tiring", and "I did not like gymnastics because I could not do a cartwheel". Conversely, four children ($n = 4/26$, 15%) reported their dislike was wanting more gymnastics, and two children ($n = 2/26$, 8%) reported no dislike as they "loved it all".

Question 3: How did you feel during the gymnastics? Mostly happy and excited.

For this question, children may have had multiple responses. Twenty-three children ($n = 23/26$, 88%) reported positive feelings during gymnastics, e.g., feeling happy and/or excited. Six children ($n = 6/26$, 23%) identified a physical response to the activity and this included feeling tired, sleepy, hot, and/or sweaty. Three children ($n = 3/26$, 12%) reported negative feelings during gymnastics and this included feeling "nervous", "tired", and "bored". It is, however, noted that the three children who expressed negative feelings for this question did include positive comments for the first question, as they identified that they liked the "bars", "the trampoline because it made me jump so high", and "the trampoline because it was fun jumping", respectively.

Examples reproduced from the children's worksheets

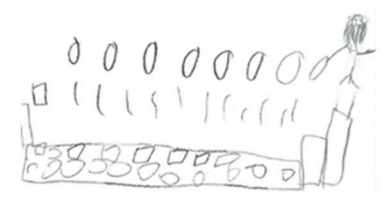
How did you feel during the gymnastics? Bored and Happy and tired and excited	How did you feel during the gymnastics? Tired But Happy
How did you feel during the gymnastics? happy sweaty	How did you feel during the gymnastics? I felt good. It was exciting
How did you feel during the gymnastics? I felt hot But happy	How did you feel during the gymnastics? it was easy and hard
Draw a picture of yourself at gymnastics: 	Draw a picture of yourself at gymnastics: me doing the Rings 
Draw a picture of yourself at gymnastics: 	Draw a picture of yourself at gymnastics: 

Figure 2. Reproduced examples of responses to children's written worksheet.

Pictorial results: 'Draw a picture of yourself at gymnastics'.

All 26 children ($n = 26/26$, 100%) drew a picture of the gymnastics program (examples of the pictures are presented in Figure 2). Twenty-one children ($n = 21/26$, 81%) drew a picture of themselves participating in the gymnastics program, and these pictures included the trampoline, bars, beams, ninja climbing equipment, and the rings. Of these 21 drawings, six ($n = 6/26$, 23%) included other children. The remaining five children ($n = 5/26$, 19%) drew a picture of themselves, smiling, without any gymnastics equipment.

Thematic analysis of children's written responses.

Across the three written questions, the theme of 'individual preferences' emerged. For the 26 children, there was a diversity of responses that indicated that the children had individual experiences while participating in the gymnastics program. One example was the bars, with some children stating they liked "the monkey bars because it was a bit scary" and because they liked being "upside down", whereas other children said they did not like the bars "because it is scary being upside down" and "because it was too high" (Figure 2).

Across the three written questions, the theme of 'resilience' emerged. In the current study, resilience was defined as a child's capacity to adapt positively to a challenge [24,25], and a child was considered to show resilience when they reported feeling positive about the program, despite identifying a part of the program they felt challenging or disliked. Across the three written questions, a theme of resilience was identified for 13 children ($n = 13/26$, 50%). Examples include a child who reported that they felt excited during gymnastics despite feeling scared when upside down on the bars; a child who reported that gymnastics made them feel both nervous and happy; a child who reported that they felt excited during gymnastics despite not being able to cartwheel; a child who reported that they felt happy during gymnastics despite not liking the floor; and a child who reported that, while on the beam, their knee was hurting—however, they continued to participate and that they "loved monkey bars and the rings and the front flips and handstands".

3.3. Experience of Gymnastics Coaches, Students on Placement, and Teachers of the Gymnastics Program (Group ii)

Across the three focus groups, the following themes emerged: (i) the impact of the program on the children's participation, (ii) the impact of the program on coach, student, and teacher attitude, (iii) the impact of the program on coach, student, and teacher knowledge, (iv) unintended impacts, and (v) barriers, facilitators, and suggestions for next time.

Theme (i): Impact of the program on the children's participation (Box 1).

Coaches reported they were able to cater to children across a range of abilities, and this individual attention enhanced participation. Program modifications week to week were found to enhance participation, to mirror each child's abilities as they improved.

Students reported they observed increased participation by the children week to week, with a reduction in the amount of self-directed "time out" during the rotations, and an improvement in their physical skills displayed during the program. Including a fine motor skill activities box in each rotation from the student's perspective meant that children could participate in fine motor activities if they were too tired for the gross motor gymnastics activities.

Teachers reported having access to the world-class gymnastics facilities, with an abundance of activities, and this enhanced the participation of the children. Adapting the program to each child individually built resilience in the children, and the teachers saw week-to-week improvements in the children's participation.

Finally, the coaches, students, and teachers all noted that, as the children became more familiar and confident, the children's preferences were sought and the children were given greater choice with and control over the individual activities, and this resulted in greater participation (i.e., children spent less time wandering away or having a "time out" during the session).

Box 1. Quotes for the impact of the program on the children’s participation.*Coach quotes*

“So, we’re catering for children with low confidence or [limited] skill to those who are actually quite able, and were able to do that quite seamlessly, which is a big credit to the coaches, with their expertise.” (Coach Participant 1)

Student quotes

“Those tiny little progressions . . . even across a couple of weeks, you can make such an impact.” (Student 7, Physiotherapy Student)

“I think having a place where they can be inclusive is really helpful for that just to build their social skills, their physical skills, as well as just . . . life skills as well.” (Student 6, Physiotherapy Student)

Teacher quotes

“You built resilience in the kids, like, you know, they fall, but they kept trying, in a very fun way.” (Teacher 1)

“For me, it was also seeing confidence from, you know, kids that were scared, just from a low beam or, you know, never tried anything or hanging off something, you know, holding on your hand. And then the next week, they go and didn’t have to hold their hand anymore.” (Teacher 2)

“And providing them with an opportunity to do something that they may otherwise not be able to do. And also the school community parents knowing that that’s available. And that’s something that, you know, the children got a lot out of and enjoyed.” (Teacher 5)

“I really liked it, because I thought that every child was catered to, you know, there was an abundance of activities for them to do. And you know, no matter what if they were, you know, had really good skill with something and not so much in the other. Like there was just activities for all. So I feel like especially my class was really catered to. We were probably lucky in the junior section that probably 100% of the students wanted to go, got on the bus got there and enjoyed it.” (Teacher 1)

Theme (ii): Impact of the program on coach, student, and teacher attitude (Box 2).

Coaches reported that the program positively impacted their attitude on hosting a program for children with a disability. Prior to this program, coaches noted that they lamented the fact that they did not have a comprehensive program for children with a disability, and that they had concerns about the children’s safety and access to the equipment. Coaches noted that being a part of this program was a positive experience that made them feel happy and made them want to continue the program. They also felt more comfortable and confident working with children with a disability.

While students reported feeling joy at seeing the children progress with their skills during the program, they also reported feeling overwhelmed at times due to the complexity of working with a group of children with diverse disabilities. Teachers reported feeling joy at seeing the children achieve skills in the gymnastics program.

Finally, across the three focus groups, all participants reported feeling more comfortable and confident within themselves around people with a disability and using the word “disability”. As the focus group participants felt more comfortable and confident during the gymnastics program, they reported that they were able to give the children more choice and control.

Box 2. Quotes for the impact of the program on coach, student, and teacher attitude.*Coach quotes*

“And this project has been an exact demonstration of how we can do that. Not only like you guys have mentioned for the children themselves and how incredible it is, and really helping with their foundations of movements and all the amazing things that come along with that. But for what it’s done for our club and our coaches, as well. I think it’s given everyone a really great sense of gratitude to purpose and definitely left us all with a big smile on our faces.” (Coach Participant 2)

Box 2. *Cont.*

“What I feel is joy, that we are able to provide this opportunity that makes me really happy. There’s also a little bit element of relief. And I say that because we’ve long lamented the fact that we have not had an inclusion program here previously. That’s not true, we’ve had inclusion support, but not have the program we really wanted to have. And the biggest reason for that is just finding people that will drive it. . . .But having someone that will really advocate to us and say we need this is what we need to be inclusive. And so that’s been a really a really wonderful thing. I particularly love the parents coming on the fifth week, I think it’s absolutely fantastic that they get to be a part of that.” (Coach Participant 2)

“I felt like, I’ve come a long way, personally, just being a part of this program.” (Coach Participant 4)

“I was apprehensive around the word disability, well not the word, but like everything that comes with it, because I didn’t know as much as I do now. But yeah, now when I hear it, I just think oh, just like bit of extra support, check in at the start, see what they need, and then just continue basically.” (Coach Participant 5)

Student quotes

“I’ve seen a slight change to what a disability may or may look like, I’ve seen a whole variety of people who may define themselves with the word disability, and to see how each of them is so individual and have different assets and things that they want to focus on work on. That that’s opened my eyes to the definition of disability.” (Student 4, Community Support Services Student)

“I absolutely loved watching the blossoming of the kids. I think watching their level of understanding and then grasping an idea of a simple tuck jump, or star jump is just awesome. Like, it’s just cool.” (Student 5, Community Support Services Student)

“I feel at times, it was a bit overwhelming. Especially when the kids don’t listen to you. I feel like . . . [it] really, really puts it into perspective, when you have a great class where everyone listens. And then you have a class where everyone’s running away or doing their own thing, it really shows you that there is up and downs to how you feel out in the gymnastic centre. But I feel like mostly it’s enjoyment, because I feel like after you finish doing something, like working with children of any kind, you kind of feel nice afterwards because you get to see how happy they are and their achievements. And I feel like it’s always you feel like you’ve done something good at the end of the day.” (Student 7, Physiotherapy Student)

“You felt really good because you could see the changes just in those five sessions.” (Student 5, Community Support Services Student)

“Within the Disability Support sector, I feel like there’s a lot of discussion within genuine inclusion, and communities. And I feel like this is definitely a big step towards that genuine cohesion within the mainstream gymnastics community.” (Student 2, Disability Support Student)

Teacher quotes

“I certainly experienced so much joy, just watching the, you know, the students being able to achieve, you know, what they didn’t think that they could do, there was so much opportunity there. And to have that opportunity, we wait to keep trying and to keep improving, or even just have a go and get over those anxieties.” (Teacher 8)

“It was brilliant, loved it, love seeing all the kids, at their own levels, enjoy what they felt comfortable doing, and confident enough to do.” (Teacher 1)

Theme (iii): Impact of the program on coach, student, and teacher knowledge (Box 3).

Coaches reported that working with children with a disability helped them to develop a new way to approach gymnastics coaching, and a new way to communicate and connect with the child. The coaches reported that the skills learnt in this program were translatable to the mainstream gymnastics program where the individual needs of the children could be taken into account, rather than a one-size-fits-all approach for the whole class.

Students reported that working with a multidisciplinary team, that included the teachers, coaches, and students from different disciplines, provided an opportunity to brainstorm how to best provide care for each child, to enhance their experience. Students felt that the improved communication skills acquired would be transferable to other settings.

Teachers reported that they were able to impart knowledge (e.g., summary of the children prior to the program commencing), and enhance the coach and students’ communication skills while on placement.

Box 3. Quotes for the impact of the program on coach, student, and teacher knowledge.*Coach quotes*

"It's just a different way of thinking about things and realising that all children you can connect with and all children learn differently. So I'm not for me, not necessarily around the language, but just more so learning styles and how that can translate to every child, as opposed to just those children with a disability." (Gymnastics Coach 5)

"We got to learn about visual cues and colours, and it makes it more fun to explain things as well. And I now do that in my normal classes as well, and it's working a lot better. So, I learned a lot in that way from a coaching perspective." (Gymnastics Coach 4)

Student quotes

"I really enjoyed sort of coming together and speaking about the children in learning about who they are, and having so much input from so many people on one child, I feel like it's really special, because they might not always get that individualised treatment in their everyday life. So the fact that we can all come together and brainstorm an idea about one child about how we can improve their experience or make them feel more comfortable." (Student 3, Physiotherapy Student)

"I do think that a lot of the skills that we've developed and learned in this placement are transferable. So I think a big one would be communication and especially in you know, in sport and in this sport, having to develop those different communication techniques, the verbal the nonverbal demonstration, the feedback is a very transferable skill into all your interactions with anyone, regardless of you know, ability or disability." (Student 1, Physiotherapy Student)

Teacher quotes

"And the coaches and students also said, having those summaries on each child was just amazing to, you know, to go in there with a little toolbox for each child of what works, what doesn't work, and build on that kind of thing. So that was really, really good two way communication." (Teacher 7)

"In the beginning, the instructors were using too much language, instead of just keeping it short and succinct. But when they're demonstrating that we're turning their backs and still talking this and I'm thinking you've lost the kid, so not even, you know, so. But that was just fine tuning in the beginning." (Teacher 3)

Theme (iv): Unintended impacts. (Box 4)

Coaches reported one unintended impact was that learnings from this program have been translated into the mainstream gymnastics program. For example, one coach noted that they were able to connect with and teach each of the children with a disability, by modifying their teaching style, and that this skill was helping them teach the mainstream classes.

Students from multiple courses reported being more open to a career that included working with people with a disability, following participation in the gymnastics program.

Teachers reported there were additional benefits for the children, such as turn-taking and developing social skills. The carryover of activities and skills learnt in the gymnastics program into the school playground and community playgrounds while on school excursions was also observed by teachers.

Box 4. Quotes for unintended impacts.*Coach quotes*

"One thing for me and not necessarily so much around the language side of things, but more so around how transferable the skills are, in terms of coaching, mechanisms, strategies, techniques, they are very similar, and if not the same as what we would do it in, in our standard classes. And I think even some of the stuff that the coaches have learned and some of the stuff that that we've all discussed, is very applicable, and sometimes even more beneficial for our standard educational classes. It's just a different way of thinking about things and realising that all children you can connect with and all children learn differently. So I'm not for me, not necessarily around the language, but just more so learning styles and how that can translate to every child, as opposed to just those children with a disability." (Coach Participant 4)

Box 4. *Cont.*Student quotes

“Was really exciting to me to see how I could possibly include into my future career and just continue to make more of an impact in the disability space” (Student 7, Physiotherapy Student)

“I’m able to sort of take that into my future career of working with a different population or population that I wasn’t expecting to be exposed to or work with. Before starting this.” (Student 4, Community Support Services Student)

Teacher quotes

“And for me, I think the bonus was teaching them how to take turn it using social skill, the aspect of it, yes, it was really helpful.” (Teacher 4)

“You had the gymnastics program. But then, as was already mentioned, you’ve got all that the independence functional skills, you’ve got the socialization, all those personal and social skills that came into it as well. Yeah, some of our kids are still talking about now.” (Teacher 6)

“We went to the park and they kids used the balance beam like at gymnastics.” (Teacher 2)

“We started using the gymnastics songs at assembly, and they really enjoyed it.” (Teacher 7)

“I had a child who has CP, cerebral palsy, and he is very scared of going on the [school] equipment. But then ever since the gym program, like when he’s back at school, like he gives everything a try.” (Teacher 1)

Theme (v): Barriers, facilitators, and suggestions for next time (Box 5).

Coaches reported the following: program facilitators were having the gymnastics space dedicated to this program to reduce distractions; the high ratio of adults to children (1:2); having a highly organised program that could adapt week to week; and “there is something for everyone to achieve within the program” (Gymnastics Coach 5). The inclusion of ‘open week’ during week 5 of the program where the parents, carers, and friends were able to come and watch the program was a facilitator, as this connected the families to the school-based program. The program barriers reported were concerns over children’s safety; however, it was noted that these concerns were alleviated by safety strategies put in place, for example, higher adult-to-child ratios and activities that could be modified to meet the needs of the individual child. Coaches suggested that, for future programs, previous years’ coaches could provide insights into the program to reduce future coaches’ concerns that relate to working with children with a disability. In terms of duration, coaches felt that future programs should run for full days, not just 2 h, to make it financially worthwhile to commit to coaching in the program.

Student-reported program facilitators were as follows: the opportunity for paediatric experience, and working with a multi-disciplinary team; having the teachers involved as they already knew the children well; and having the children participate with their school-friends to support one another. A reported program barrier was having the program spread over 10 months (3 h per week), rather than a more intensive period of full days, for practical reasons. Having parents attend in week 5 (‘open week’) was seen as both a facilitator and a barrier, as some children were excited and pushed themselves with the family present, whereas others were distracted or stepped back with their family present.

Teacher-reported program facilitators were as follows: having the program manager spend two days at the school to meet the children and learn the school routines so they could be embedded into the gymnastics program; the consistency of coaches and students and the length of the program (5 weeks), as this enabled the children to learn the new gymnastics routine; having the gymnastics centre exclusively; using a bell to signal a change of rotations; the safety strategies that were put in place such as locking the inside gymnastics doors to ensure the children remain in a safe space; and having access to fine motor activities within each rotation, which helped children to stay engaged when they needed to rest. A program barrier reported was the times the gymnastics space was shared with other gymnasts, as this could distract the children. Teachers suggested future programs could build on the facilitators as reported above, such as ensuring the consistent exclusive use of the gymnastics centre, extending the program beyond 5 weeks, and ensuring the kids have a dedicated drink break during each session.

Box 5. Quotes for barriers, facilitators, and suggestions for next time.*Coach quotes*

"At week five their parents get to come and watch, such a collaborative moment so that those families can see what they have achieved within that time, I think [that] has been really positive." (Gymnastics Coach 2)

"Everything was very, very well organised. The coaches knew where they were going. They knew things about their children, they knew the children's names, they followed the rotations, the bell rang on time. And this just set the children up for success." (Gymnastics Coach 3)

"Each week, we've learned from the previous week and implemented the new things that we need moving forward to ensure we are getting the best out of the children, and that the children really learn something." (Gymnastics Coach 1)

Student quotes

"I do think that what may be more beneficial is to have all that right at the start before we went into the gymnastics program. So going to the special school and meeting the teachers and learning that side and having more of an in-depth workshop about what the program is and the grant process and just how it all came about at the start of the process of the sessions instead of maybe in the middle, because I feel like a lot of us came in not knowing what to expect." (Student 5, Community Support Services Student)

"I think I would still like to be involved in the process. However, the actual hours and placement per se, I would say that, I probably wouldn't want to continue it next year. And my only reasoning is because it's five weeks out of each term. And it's three hours in the middle of the day, which is a lot of time to take out every five weeks." (Student 2, Disability Support Student)

"I get so much joy out of working with the kids on the floor. And then coming up here [board room] having group discussions, working with people, and then learning their insights opens and broadens my mind as well. So I gained a lot from that too. But then again, from physio lens, we don't have that many opportunities to work with kids, or especially, you know, kids with disabilities." (Student 7, Physiotherapy Student)

"I think that having families come in is a facilitator and a barrier in some ways, depending on the child. And then some children are getting quite distracted, trying to walk right up to their parents or trying to get their attention away that then and but when we know that they can be way more engaged, or they have been way more engaged in previous sessions or previous weeks. And then for other students, it was . . . [motivating to have] the parents . . . supporting them, cheering them on. They were performing, you know beyond what we've ever seen them during the past few weeks." (Student 2, Disability Support Student)

Teacher quotes

"Our kids learn through routine and it can take them a long time to get used to change. . . So to have it any shorter than five weeks, we wouldn't have gained as much out of it." (Teacher 4)

"100% I think it's such a fantastic opportunity for our kids. And yeah, just giving them that experience is something that yeah, I would definitely be involved in again, for sure. Probably the thing that I liked the most out of it was the length of time or given to get into the program and then to see the kids actually develop core strength and skills, has been really terrific." (Teacher 1)

4. Discussion

The children with a disability were able to clearly identify what they did and did not like about the gymnastics program ('George's Gymnastics Program'). All children identified a part (or multiple parts) of the program that they liked, and most children reported that they felt happy and excited during the program. However, there was diversity in the responses, and some contrary responses, such as one child who reported they liked being upside down on the bars and another child who reported that they did not like being upside down on the bars. While most children participated in the full 1 h session, a small number of children wandered away or had a "time out" during the session. As the coaches, students on placement, and teachers became more familiar and confident, the children's preferences were sought and they were given greater choice with and control over the individual activities, and this resulted in greater participation (i.e., children spent less time wandering away or having a "time out" during the session). This indicates that, while this gymnastics program for children with a disability was well-liked by the children, future

programs need to allow for individual children's preference and promote individual choice and control, to maximise participation in the program.

In our study, the themes of resilience as well as individual preferences and the tailoring of the program emerged from the children's written responses, and from the focus groups of the adults who supported the program. Children reported, and were observed to show, resilience and adaptability during the gymnastics program. In our study, tailoring was seen to be an important facilitator to both participation and enjoyment, and resilience. Adults observed that children appeared to enjoy participating in structured physical activity, and participation was enhanced when the child was given choice and control. Previous research has also shown that tailored, accessible, and supportive environments make a difference in gymnastic participation by children with a disability [13]. In addition, coaches' knowledge of how to engage children with a disability in gymnastics positively influenced the children's ability to have choice and control within the program, with the gymnastic coaches' knowledge viewed as important by parents [13]. Our results suggested that it is possible that the positive experience of physical activity in this setting can support the desire for further participation in physical activity, for children with a disability. Barriers to physical activity participation for children with a disability include practical limitations such as program and staff capacity, time constraints, the child's preferences, parental behaviour, an unwillingness to be inclusive, negative attitudes towards disability, and a lack of skilled instructors and opportunities [6–8]. This is in contrast to the facilitators to physical activity participation for children with a disability which include the child's desire to be active, the involvement of peers, family support, programs that promote success and inclusion through fun and inclusive activities, programs that have skilled staff and a pathway as children grow and develop their skills, and programs with strong partnerships across the disability, sport, education, and government sectors [6–8].

In our study, it was apparent that this health promotion program positively impacted the coach, university/TAFE student, and teacher attitude and knowledge. Sports clubs are important for health promotion; however, there is significant organisational change required to embed health promotion into sporting club program and culture [26]. While inclusive community sports programs benefit children with a disability, access is limited, and the children's preferences for these programs are largely unknown. A recent study found that gymnastics had physical and social benefits for children with a disability [13]. However, addressing gaps to support children's access to gymnastics is important. Gaps may include children with a disability having reduced access to inclusive community gymnastics programs, and that gymnastics coaching shortages limit the delivery of inclusive community gymnastics programs [13]. This current study addresses these gaps by improving access while concurrently focusing on the student voice, where children with a disability were encouraged to communicate their ideas and opinions about participating in an inclusive community gymnastics program (via the evaluation). This, in turn, gives students the power to directly influence change on the next iteration of this inclusive community gymnastics program, as well as other programs that adopt the findings of this study.

This study has several strengths and limitations. The strengths included a five-week program that was repeated over four school terms, which allowed time for reflection, modification, and improvement; as well as an evaluation that included the four groups of key stakeholders, namely, the children, gymnastics coaches, students on placement, and teachers. The limitations included only including children from the special school's Senior section (11–12 years), meaning that the findings may not be generalisable to the children from the Foundation (5–6 years), Junior (7–8 years), or Middle (9–10 years) sections, as well as results that are limited to children with a mild intellectual disability. There was also potential bias in the children's data collection, in the way the instructions were provided to the children, despite the research team's best effort to reduce bias by pre-testing the worksheet and collecting data under the supervision of the English Specialist teacher who had not been involved in the gymnastics program. We have identified a number of

facilitators and barriers to support the successful implementation of a gymnastics program for children with a disability. These insights have allowed for the reflection of elements that should be included and any adaptations that should be considered in future iterations of the gymnastics program for children with a disability (Appendix B). Future research in this domain should build on this study examining participant attitudes, by measuring outcomes and using a randomised design, to understand the impact of the program. To maintain the anonymity of the small group of children participants in this study, demographic information regarding disability diagnosis was unable to be provided as part of the results.

5. Conclusions

While ‘George’s Gymnastics Program’ for children with a disability was well-liked by the children who participated, and positively benefited the adults who supported the program, future programs need to promote individual children’s choice and control, to optimise participation. The next iteration of this program should address the barriers identified, and future research in this domain should go beyond examining participant attitudes, by measuring outcomes and using a randomised design, to understand the effect of the program on physical, psychological, and social outcomes.

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Institutional Review Board Statement: This study was approved by the Monash University Human Research Ethics Committee (Project ID #38902; approved 18-07-2023).

Informed Consent Statement: Informed consent was obtained from all participants involved in the study.

Data Availability Statement: Upon reasonable request, deidentified participant data are available from Natasha K. Brusco.

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Appendix A. Topic Guide for the Focus Groups

IMPACT

- What did the inclusion gymnastics program mean to you?
- How did participation in the inclusion gymnastics program impact you:
 - (a) Professionally?
 - (b) Personally?
 - (c) In other ways?

- Did participation in the inclusion program change the level of comfort you feel when using the word “disability”?

ADULTS REACTION TO THE CHILDREN’S DATA COLLECTION QUESTIONS (in reference to themselves):

- What did you like about the gymnastics program?
- What didn’t you like about the gymnastics program?
- How did you feel during the gymnastics program?
- Given the opportunity, would you be involved in the gymnastics program again?

LOCAL ENVIRONMENT MODEL (incorporating the International Classification of Functioning, Disability and Health (ICF) levels) [19]:

- Reflect on barriers and facilitators to the inclusion gymnastics program in the context of:
 - Safety, social connections, family-centredness, and learning-based aspect;
 - Choice, fun, friends, specialised health professionals, and time;
 - Achievement, aspiration, friends, independence, enjoyment in physical activity, body function, and activity levels.

Appendix B. Proposed Changes for Future Iterations of the Gymnastic Program

Elements of the 2023 Gymnastic Program, Including Implementation	Proposed Change for the Future Iteration of the Program
Engagement of past coaches to provide insights into their participation of the program for future coaches.	A one-off mentoring session will be offered prior to the commencement of the program to provide an opportunity for future coaches to gain insights into the program.
Increasing the length of the gymnastic program to be a full day (to increase financial viability for coaches and for student placement hours to be more concentrated over a shorter period of time).	Restructure of the program timetable to provide multiple sessions on the one day.
Introductory day/session for gymnastic coaches and students on placement prior to the program commencing to improve handover of information about children participating.	As part of the ‘orientation’ to the gymnastics program, a short session will be included to allow for handover of information from parents and/or teachers.
Have the program manager spend two days at the children’s school to understand the school routine and meet the children who will be included in the gymnastics program.	This element was seen as a facilitator in the current program, and should be included in future iterations of the program.
Exclusive use of the gymnastics centre.	Future programs to have exclusive access to the gymnastics centre during the program.
Extending the program beyond 5 weeks, while maintaining consistency of coaches and students on placement.	Future programs will be of 10-week duration (duration of a school term).
Dedicated drink break embedded during session plan when establishing gymnastic routine for children.	For future session plans, rest/drink breaks to be formally embedded into the structure.
Use of a bell to signal a change in activity rotations.	This element was seen as a facilitator in the current program, and should be included in future iterations of the program.

Elements of the 2023 Gymnastic Program, Including Implementation	Proposed Change for the Future Iteration of the Program
Creating a safe space for the children, e.g., locking of gymnastics door to ensure children remain in the gymnastic centre.	This element was seen as a facilitator in the current program, and should be included in future iterations of the program.
Activity choice, including fine motor activities, to be made available to children during each activity rotation.	This element was seen as a facilitator in the current program, and should be included in future iterations of the program.

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